

Question Paper

Exam Date & Time: 25-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH. EXAMINATIONS - NOVEMBER 2025

High Performance Computing Systems [CSS_5103]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Evaluate the trade-offs between using parallel for and parallel sections. Develop a OpenMP program that uses reduction clause to calculate the dot product of two vectors of size N. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Illustrate how digital computers are classified in Flynn's classification along with their applications? Also, mention on what basis they are classified (3)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) Explain parallel processing ? Describe 3 types of concurrent events. (2)
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- 2) Develop a MPI program to read a value M and N X M number of elements in the root. Using N processes do the following task. Find the square of first M numbers, find the cube of next M numbers and so on. Print the results in the root. Do not use any built-in math functions. (4)
- A) Course Outcome: "CO3" / Cognitive Level: "Create"
- B) Implement the Odd-Even Transposition Sort algorithm for sorting N elements using a one-dimensional mesh processor array. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- C) Analyze how MPI_Reduce() can be used to perform a global sum or maximum operation across all processes. Explain how reduction improves performance compared to local computation and communication. (3)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 3) What is Command queue ? Illustrate the OpenCL steps required to transfer data from the host to the device, execute a kernel, and retrieve the result. (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Application"
- B) Construct an OpenCL kernel for multiplying two matrices A and B of size M X N and N X P. Each row of A has to be multiplied by a separate work item. (3)
- Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Determine how OpenCL can be used to implement a program that utilizes both CPU and GPU for parallel computation. Explain how this heterogeneous execution impacts the performance of the application (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4) Construct a parallel algorithm to sum n values using the Hypercube SIMD model, where n is the total number of values to be added and p is the number of processors. (4)
- A) Assume that each processor initially holds one of the n values.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Illustrate the working of a Distributed Memory architecture in the multiprocessor by describing how data is shared among processors. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Discuss the taxonomy and classification of static and dynamic interconnection networks used in parallel computer architectures. (3)

Course Outcome: "CO6" / Cognitive Level: "Understand"

- 5) Develop a CUDA program to read a matrix of size 3×3 . The kernel you write should meet the following specifications: (4)

- A) With respect to each row in the matrix, find first element to the power of 2, second element to the power of 3, third element to the power of 4. An example data is shown in Fig. Q 5A. You need to use number of CUDA threads as number of rows of matrix to make this task feasible.

Matrix A =

1 2 3
4 3 2
3 2 2

Matrix B =

1 8 81
16 27 16
9 8 16

Course Outcome: "CO5" / Cognitive Level: "Create"

- B) Demonstrate various types of CUDA program variables and their memory types, scope and lifetime. What happens when we use `__host__` and `__device__` in a function declaration? (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Develop a CUDA kernel for matrix multiplication in which each block computes one row of the resultant matrix (3)

Course Outcome: "CO5" / Cognitive Level: "Create"

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Event Name	Subject
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058-END SEMSTER-ODD-2025-PG CSS_5103:High Performance Computing Systems

#	Course Outcome	Description	Marks
1	1	CO 1	12
2	2	CO 2	5

3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
6	6	CO 6	3
Total			50

* NA - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	5(10%)
3	Application	8	26(52%)
4	Analysis	2	8(16%)
5	Create	0	0(0%)
6	Evaluate	3	11(22%)
Total		15	50 (100%)